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Comparison between hamstring autograft and free tendon Achilles allograft: minimum 2-year follow-up after anterior cruciate ligament reconstruction using EndoButton and Intrafix

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Abstract

Purpose This study is to compare the clinical and radiographic results of anterior cruciate ligament (ACL) reconstruction with four-stranded autogenous hamstring tendon and two-stranded free tendon Achilles allograft fixed with EndoButton in the femoral tunnel and Intrafix in the tibial tunnel.

Materials and methods 106 patients diagnosed with ACL rupture underwent ACL reconstruction. Autogenous hamstring tendon was used in 33 patients (group I) and free tendon Achilles allografts were used in 32 patients (group II). Median age was 23 years old (20–51) in group I and 22 years old (20–55) in group II. Range of motion, Lachman test, Pivot shift test, IKDC score, Lysholm score and side-to-side difference (SSD) were evaluated preoperatively and at the last follow-up. Tegner activity scale was evaluated before injury and at the last follow-up.

Results The mean follow-up periods were 28.1 months in group I and 31.6 months in group II. Range of motion of the knee was not different from that of the unaffected side in most cases except one flexion deficit in group I and three in group II (n.s.). One in group I and three in group II showed grade two or three laxity on Lachman test at the last follow-up. One in group I and three in group II showed clear positive results on Pivot shift test at the last follow-up. Thirty in group I and 26 in group II were classified to IKDC A or B at the last follow-up (n.s.). Median Lysholm scores were 98 (85–100) in group I and 99 (85–100) in group II at the last follow-up (n.s.). Median Tegner activity scales were 6 (5–9) in group I and 6 (4–9) in group II at the last follow-up (n.s.). The mean SSD at the last follow-up were 1.4 ± 2.0 mm in group I and 1.9 ± 2.4 mm in group II (n.s.).

Conclusion Clinical and radiological outcomes of ACL reconstruction with two-stranded free tendon Achilles allograft were comparable to those of four-stranded autogenous hamstring tendon. This technique is reasonable to accomplish good results without some weaknesses when using allograft with bone block.

Level of evidence Therapeutic randomized controlled prospective study, Level I.

Keywords Anterior cruciate ligament · Hamstring · Achilles · EndoButton · Intrafix

Introduction

There are several kinds of grafts for anterior cruciate ligament (ACL) reconstruction and, among them bone-patellar tendon-bone (BPTB) and hamstring are generally used as autografts and Achilles tendon and BPTB as allografts. Achilles tendon allograft generally has bone block on one side and free tendon on the other side. It is usually fixed by bone-to-bone fixation in the femoral tunnel and by tendon-to-bone fixation in the tibial tunnel. The advantages of the Achilles allograft over Hamstring autograft are no pain in the donor site and bone-to-bone fixation which is

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generally more firm than tendon-to-bone fixation [3, 22]. The bone blocks, however, sometimes may crack during operation, making firm fixation difficult. The authors reduced the risk of loosening caused by bone block cracks and saved more host bone by using free tendon Achilles allograft instead of Achilles allograft with bone block on its end. Achilles allograft was prepared to be looped double strands and was fixed in tunnels without bone block. The purpose of this study is to compare the clinical and radiographic results of ACL reconstruction with autogenous hamstring tendon and free tendon Achilles allograft fixed with EndoButton in the femoral tunnel and Intrafix in the tibial tunnel. Null hypothesis was that the outcomes of ACL reconstruction with free tendon Achilles allograft restores are equal to those of autogenous hamstring tendon.

Materials and methods

In this randomized controlled prospective study of consecutive series, all subjects were informed of the study procedure, the purpose of the study, and any known risks; and all provided informed consent. The study was conducted with the approval of the Ethics Committee of National Police Hospital in Seoul, South Korea.

One hundred and ninety-three consecutive patients were diagnosed with ACL rupture and underwent ACL reconstruction by a single surgeon between 2006 and 2008. The patients who underwent a subtotal or total meniscectomy, meniscal repair, or meniscal transplantation due to a meniscus injury, who underwent an operation due to any concomitant ipsilateral ligament injuries, who had histories of any ligament injuries on contralateral side, or who underwent microfracture or cartilage transplantation due to osteochondral lesions were excluded. Seventy-four participated in this study and sixty-five of them were followed up for at least 2 years after the operation. Autogenous hamstring tendon was used in 33 patients (group I) and Achilles allografts (LifeNet Health Inc., Virginia Beach, VA, USA) were used in 32 patients (group II). The allografts were fresh frozen at −80°C and thawed at room temperature just before the surgery. Group I consisted of 30 males and 3 females, while group II consisted of 26 males and 6 females. The median age was 23 years old (20–51) in group I and 22 years old (20–55) in group II. The numbers of patients who underwent an operation within 3 months after injury were 29 in group I and 28 in group II. The mechanisms of injuries are described in Table 1.

Diagnosis of ACL rupture was made by physical examinations and MRI and confirmed by arthroscopy. Lachman test and pivot shift test were performed under anesthesia. Tip of the tibial tunnel guide was placed at 8 mm anterior to the posterior cruciate ligament. The guide pin was inserted with an angle of 45° to the tibia. Then we made certain that the anterior cortical bone anterior to the guide pin was sufficient using curved hemostat to avoid the rupture of the anterior cortical bone during reaming. The diameter of the tunnels was the same as that of the graft. The femoral tunnel was prepared via a transtibial technique. It was drilled at 10:30 in the right knee or 1:30 position in the left knee with 1.5–2 mm thickness of the posterior cortical bone remaining.

In all the patients where autogenous hamstring tendons were used, the ipsilateral semitendinosus and gracilis tendons were harvested and the grafts were doubled to become four strands. In patients where Achilles allografts were used, Mayor Scissors and No. 10 blade were used to harvest 5–7 mm wide free tendon strip after Achilles allograft had been thawed. Free tendon strip was folded into two strands and its thickness became 8–9 mm. The length of the graft was made to be 9.5–10 cm long and whipstitches were conducted on each strand of tibial end 35 mm using no. 2 ethibond while maintaining the space between each stitch constantly at around 5 mm. 30–35 mm of looped end was inserted into the femoral tunnel and fixed by an EndoButton CL (Smith & Nephew, Endover, MA, USA). After pretensioning was achieved by conducting 20 range-of-motion cycles under maximal manual load, tibial end of the graft was fixed using a 28 mm long Intrafix (DePuy Mitek, Raynham, MA, USA) with the knee in 20° of flexion and augmented with screw and washer. Continuous passive motion exercises from 0° to 50° began on second postoperative day and advanced to 90° on second postoperative week as tolerated. Partial weight bearing was allowed on the second postoperative day, and total weight bearing was permitted as tolerated. After 12 weeks, jogging and stationary bicycling were allowed. After 6 months, each patient was allowed to do competitive sports except for those exercises that might involve strong contacts with others such as football or soccer, or those exercises that might impose strong external forces on the patient such as skiing or snowboarding. All exercises were allowed after 9 months.

In group I, five patients had medial meniscus tears and three patients had lateral meniscus tears, and partial meniscectomies were conducted in all the cases. Four MCL injuries were conservatively treated. In group II, four patients had medial meniscus tears and three patients had lateral meniscus tears, and partial meniscectomies were conducted in all the cases. Three MCL injuries were conservatively treated.

All patients were evaluated by two orthopaedic surgeons preoperatively and 2 years after operation. As clinical evaluations, Lachman test was graded negative [0; side-to-side difference (SSD) <3 mm and hard end point], 1+ (SSD of 3–5 mm), 2+ (SSD of 5–10 mm), and

Table 1 Demographics

	Group I ($n = 33$)	Group II ($n = 32$)	*P*
Graft type	Autogenous hamstring	Allogenous Achilles	
Median age (years)	23 (20–51)	22 (20–55)	n.s.
Laterality (right:left)	14:19	12:20	n.s.
Male:female	30:3	26:6	n.s.
Trauma to operation interval (weeks)	7.2 ± 10.3	6.5 ± 8.2	n.s.
Mean follow-up period (months)	28.1 ± 6.5	31.6 ± 10.0	n.s.
Etiology			
Sports	28	25	
Fall down	2	4	
Traffic accident	2	3	
Unknown	1	0	

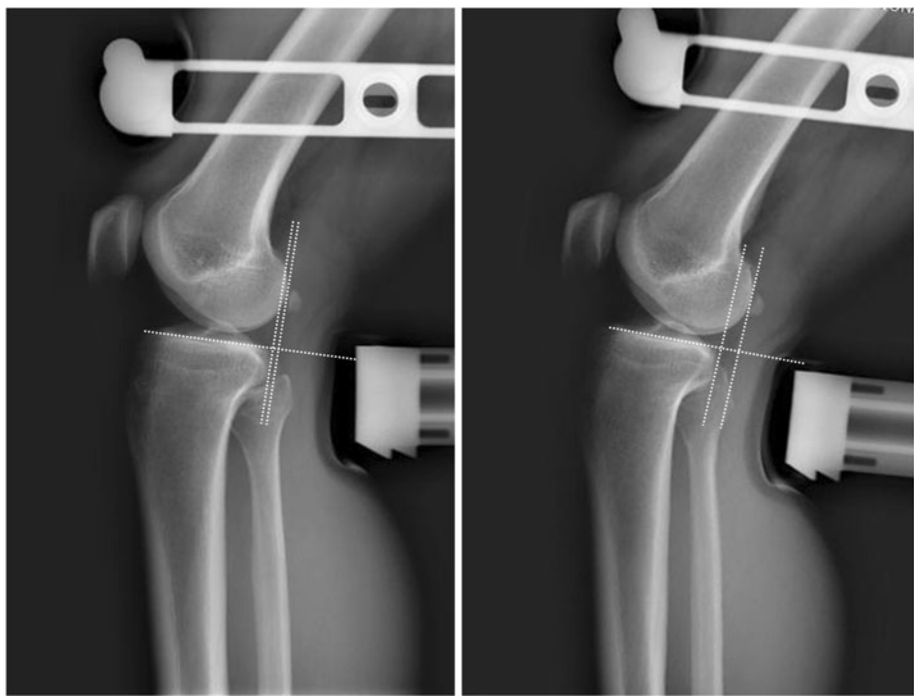

Fig. 1 Stress radiographs were taken with the knee flexion angle of 30° and the anterior load of 150N using Telos stress device

3+ (SSD > 10 mm). Pivot shift test was graded 0 (normal), 1+ (glide), and 2+ (clearly positive with clunk). Range of motion of the knee, IKDC score, and Lysholm score were evaluated before operation and at the last follow-up. Tegner activity scale was evaluated before injury and at the last follow-up. In addition to clinical examination, stress views were taken under 150N using the Telos stress device (Austin & Associate Inc., Fallston, MD, USA) to measure SSD on the Picture Archiving and Communication System (GE Healthcare, Chicago, IL, USA) preoperatively and at the last follow-up (Fig. 1). Tunnel widening was assessed with the method of L'Insalata et al. [16] (Fig. 2). Tunnel diameters were measured on anteroposterior and lateral radiographs postoperatively and at the last follow-up. The greatest diameter of the whole tunnel was measured for the femur and the tibia. The measurements were made between the inner borders of the sclerotic margins. All measurements were corrected for magnification. Measurements on radiographs were performed by two observers and they were repeated two times at 2 month intervals.

Statistical analysis

The sample size of 32 reconstructions provides 98% power at a 0.05 significance level to detect 3 mm or more of SSD. The statistical analyses of differences in continuous variables between two groups were performed with unpaired Student *t* test and differences in categorical variables with χ^2 test. To test intraobserver and interobserver reliability, the intraclass

Fig. 2 Examples of radiographs of tunnel diameter measurement

correlation coefficient (ICC) was used with the respective 95% confidence intervals. All analyses were performed with SPSS version 17.0 (SPSS Inc., Chicago, IL, USA).

Results

The mean follow-up periods were 28.1 ± 6.5 months in group I and 31.6 ± 10.0 months in group II (n.s.). The mean diameters of reamer were 7.7 ± 0.5 mm in group I and 8.6 ± 0.5 mm in group II. The interobserver ICCs for Lachman test, Pivot shift test, SSD, and tunnel diameter were >0.8, and intraobserver ICCs for SSD and tunnel diameter were >0.9. Six (18.2%) in group I and nine (28.1%) in group II showed positive on Lachman test (n.s.) at the last follow-up (Table 2). Five (15.2%) in group I and six (18.8%) in group II showed positive on pivot shift test (n.s.) at the last follow-up. The patients who showed grade II or more of Lachman test or grade II of pivot shift test were one (3.0%) in group I and three (9.4%) in group II (n.s.). The range of motion of the knee was not different from that of the unaffected side in most cases except one flexion deficit in group I and three in group II (n.s.) while none of the patients had extension deficit. Preoperative IKDC score were *C* or *D* in all cases. At the last follow-up, 56 patients (86.2%) were classified to IKDC *A* or *B* (Table 3). The median Lysholm scores were 56 (44–80) preoperatively and 99 (85–100) at the last follow-up (Table 4). The median Tegner activity scales were 7 (5–9) before injury and 6 (4–9) at the last follow-up (Table 5). The mean SSD at the last follow-up was 1.7 ± 2.2 mm (Table 6). Data of tunnel diameter and tunnel widening are in Table 7.

Table 2 Lachman test and pivot shift test at last follow-up

	Group I	Group II	*P*
Lachman test			n.s.
–	27 (81.8%)	21 (65.6%)	
1+	5 (15.2%)	8 (25.0%)	
2+	1 (3.0%)	3 (9.4%)	
3+	0 (0%)	0 (0%)	
Pivot shift test			n.s.
–	27 (81.8%)	26 (81.3%)	
1+	5 (15.2%)	3 (9.4%)	
2+	1 (3.0%)	3 (9.4%)	

Table 3 IKDC score at last follow-up

	Group I	Group II	*P*
A	23 (69.7%)	19 (59.4%)	n.s.
B	7 (21.2%)	7 (21.9%)	
C	3 (9.1%)	6 (18.6%)	
D	0 (0%)	0 (0%)	

One patient in group I had arthroscopic debridement twice due to synovitis and arthroscopic brisement due to limited range of motion. At the final follow-up, the patient showed a restriction of 5° on flexion.

Discussion

The most important finding of this study is that the outcomes of ACL reconstruction using free tendon Achilles allograft were compatible to those of autogenous hamstring

Table 4 Median Lysholm score

	Preoperative			Last follow-up		
	Group I	Group II	*P*	Group I	Group II	*P*
Lysholm score (range)	54 (44–78)	56 (45–80)	n.s.	98 (85–100)	99 (85–100)	n.s.

Table 5 Median Tegner activity scale

	Pre-injury			Last follow-up		
	Group I	Group II	*P*	Group I	Group II	*P*
Tegner activity scale (range)	6 (5–9)	7 (5–9)	n.s.	6 (5–9)	6 (4–9)	n.s.

Table 6 Mean SSD on Telos stress device

	Preoperative			Last follow-up		
	Group I	Group II	*P*	Group I	Group II	*P*
SSD (mm)	9.8 ± 4.1	9.9 ± 4.3	n.s.	1.4 ± 2.0	1.9 ± 2.4	n.s.

tendon without the risk of intraoperative complication like cracks of bone block. There are a few options in ACL reconstruction regarding graft, one being an autograft, another an allograft, and the other a synthetic which is rarely used because of high complication rate. Allograft has some advantages over autograft; avoiding donor site morbidity, reducing surgical time, smaller incisions, availability of larger grafts and no weakening of the extensor or flexor mechanism [11]. Fixation methods in ACL reconstruction are largely divided into three kinds including compression (e.g., Metal or biomaterial interference screw), expansion (e.g., Rigidfix), and suspension (e.g., EndoButton or Endopearl). In reconstruction using Achilles allograft, bone-to-bone fixation with interference screw has been conventionally used in femoral tunnel. Many studies have been reported that failure load in bone-to-bone fixation with interference screw is positive and that bone-to-bone fixation is more stable and heals faster than soft tissue-to-bone fixation [8, 13, 19, 22]. However, Achilles allograft sometimes cannot be firmly fixed due to cracks in bone plug of allograft while preparing, pulling a graft into the tunnel, fixating, or even pretensioning after femoral fixation. In addition, since the cross-section area of the attached tendon is small compared to that of the bone plug, there is a tendency that the diameter of the bone plug becomes unnecessarily big to obtain the desired thickness of the tendon. As a result, the diameter of the femoral tunnel also becomes big and this is not desirable when it comes to revision. Considering the fact that the length of the tendinous portion of Achilles allograft is generally about 18–22 cm, a graft with sufficient thickness to reconstruct an ACL can be obtained with this technique even with a small amount of graft. However, it is better to consider other fixation methods since the failure load of the interference screw in tendon-to-bone fixation is low [20, 28]. Although, Rigidfix that has been recently used has sufficient failure load and its results have been reported to be satisfactory [1, 18, 21, 27], it is controversial that there is a possibility of residual laxity by slippage [20]. In biomechanical studies on different kinds of soft tissue-to-bone fixation devices in femoral tunnels, Wu et al. [27] and Milano et al. [20] reported excellent failure loads of EndoButtons. Many studies have reported that EndoButton has a long distance from an aperture to the fixing point and thus tunnel widening may occur due to bungee effect [4, 9, 15, 25]. However, tunnel widening has been reported also in other fixation methods such as Rigidfix and interference screw and several studies have reported that tunnel widening was not related with bungee effects [14, 15]. In addition, it has been reported that tunnel widening did not affect clinical outcomes [5, 9, 12, 17]. Soft tissue fixation in tibial tunnel can be performed with interference screw, Intrafix, spiked washer screw, or staple. Although interference screw is frequently used, its fixing force is weak as mentioned before. Thus, additional fixation is performed on tibial cortex using spiked washer screw or staple in many cases. Intrafix is a method to press the graft into the bony wall using a sheath and then expand the tendon and firmer fixation is expected compared to interference screw [26]. However, it has a disadvantage that the sheath is

Table 7 Tunnel diameter

	Postoperative (mm)		Final follow-up (mm)		Tunnel widening[a] (%)		
	Group I	Group II	Group I	Group II	Group I	Group II	*P*
Femur	7.9 ± 0.5	8.6 ± 0.5	9.5 ± 0.6	10.2 ± 0.7	21.2 ± 4.3	19.3 ± 3.5	n.s.
Tibia	7.9 ± 0.6	8.5 ± 0.5	9.5 ± 0.6	10.1 ± 0.5	20.8 ± 4.9	19.1 ± 4.3	n.s.

[a] Change in tunnel diameter/postoperative tunnel diameter × 100

nonabsorbable which makes revision more difficult. When creating a tibial tunnel, the angle of the guide pin with the tibial shaft should be small to be around 45° to form a femoral tunnel using the transtibial technique. The authors experienced a case where a tibial tunnel was positioned too forward, ultimately damaging the anterior wall of the tibial tunnel. Fixation with Intrafix was too weak and thus additional post-tie was made with a washer screw. Based on the results of follow-ups, fortunately, no laxity appeared and the outcomes were satisfactory. After this experience, the bone thickness anterior to the tibial tunnel guide pin was roughly assessed by palpation using curved hemostat before reaming.

Scheffler et al. [23] reported that the healing of autografts was faster than that of allografts and although differences in histological findings between autografts and allografts decreased a year after the transplantation of grafts, biomechanical properties of allografts were inferior to those of autografts. However, the follow-up periods have been relatively short to be around 1 year in most histological or biomechanical studies. Many studies have reported that there is no significant difference in clinical outcomes comparing autografts and allografts [7, 10, 29]. In most studies, accelerated rehabilitation program is applied after ACL reconstructions without discriminating between autografts and allografts. Some adjustments of rehabilitation program may be necessary for allografts with which the healing process is slow [2, 6, 24]. In this study, the same rehabilitation program was applied to both groups and satisfactory outcomes were shown at the last follow-up and there was no significant difference in clinical and radiologic outcomes between two groups. Thus, the results support the null hypothesis.

Limitations of this study were short follow-up period, no visual identification of graft healing state with secondary arthroscopy, and unknown histologic difference. Position of the femoral tunnel is not anatomic in this study. The best position of the femoral tunnel has not been established yet. At present, anatomic position is emphasized rather than isometric position; however, exact anatomic position is debatable. Most surgeons agree that it is impossible to perfectly reproduce the anatomic ACL in double bundle as well as single bundle ACL reconstruction. The femoral tunnel tends to be more horizontal, but transtibial technique has limit to do so.

Conclusion

In conclusion, short term outcomes of the ACL reconstruction with free tendon Achilles allograft were comparable to those of autogenous hamstring tendon. This technique eliminates the risk of crack in bone block of the graft and can reduce required tunnel diameter with the same tendon volume.

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